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ABSTRACT

The field reversed configuration (FRC) has been a curious case in plasma physics research in that early MHD analysis suggested FRCs should be grossly unstable, while experimental results contradicted that prediction. Later, this theory was able to resolve this contradiction by understanding that finite Larmor radius effects largely negated the MHD predictions. Similarly, previous theoretical studies of beam driven FRCs predicted that such system would be unstable to beam driven modes while, again, experimental results indicated the contradiction. In this paper, we reconcile the theoretical understanding of beam driven modes with experimental observations of stability in these systems. By self-consistently capturing fast ion generation from neutral beam injection and its impact on the plasma equilibrium, we show that low amplitude perturbations in the magnetic field, driven by betatron particles, modify the precession frequencies of the betatron particles such that the drive for compressional Alfvén waves in the thermal plasma is reduced. Finally, we are able to demonstrate, for the first time, stable beam driven FRC evolution at high S^*/E in 3D kinetic simulations.

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I. INTRODUCTION

TAE Technologies, Inc. (previously known as “Tri Alpha Energy, Inc.” and hereinafter referred to as “TAE”) is pursuing a specific type of high- β fusion reactor approach based on the field-reversed configuration (FRC)^{1,2} (with β being the ratio of plasma pressure to magnetic field pressure). A detailed description of TAE’s approach and progress can be found in Refs. 3 and 4 and references therein. The neutral beam injection (NBI) driven FRC formation technique used at TAE is described in Ref. 5.

FRCs rely entirely on plasma current to maintain the magnetic field topology. This allows FRCs to operate at very high volume averaged β values (approximately unity). However, early evaluations of the magnetohydrodynamic (MHD) stability of compact toroids suggested that FRCs are unstable to tilt.⁶ An analysis of ideal tilting in FRCs¹ revealed a nominal growth rate of $\gamma_{MHD} \approx 2v_A/L_s$, where v_A is the characteristic Alfvén velocity⁷ and L_s is the FRC length. Critically though, elongated FRCs are observed to survive much longer in experiment than suggested by the MHD analysis. Reference 2 reports that 2/3 of the data in the FRC data compendium⁸ show particle confinement times between 21 and 63 times the MHD tilt mode growth period.

This contradiction between MHD theory and experiment is largely resolved through the inclusion of finite-Larmor radius (kinetic ion) and two-fluid (separation of electron and ion responses) effects. Several authors have reported large reductions in the tilt growth-rate observed in simulations that capture such effects vs the MHD growth rate for elongated FRCs.^{9–14} More specifically, Belova *et al.*¹⁵ found empirically that kinetic ion effects reduce the linear tilt growth rate of elongated FRCs according to

$$\gamma_{kin} \approx \gamma_{MHD} \exp(-3E/S^*), \quad (1)$$

where S^* is the ratio of the separatrix radius (R_s) to the ion skin depth and $E = L_s/(2R_s)$ is the separatrix elongation. The implication is that FRCs with $S^*/E \lesssim 3$ are significantly less unstable than ideal MHD predicts; this stability threshold was broadly obeyed by numerous FRC experiments,² including TAE’s C-2 and C-2U experiments.

The classical confinement time for an FRC is given by Tuszewski *et al.*¹⁶

$$\tau_{classic} = \left(\frac{R_s}{4}\right)^2 \frac{\mu_0}{\eta_{classic}},$$

where μ_0 is the vacuum permeability and η_{classic} is the cross-field Spitzer resistivity ($\eta_{\text{classic}} \propto T_e^{-3/2}$). Recalling the definitions of S^* and E , using $\beta = nT_i/(B^2/2\mu_0)$, and assuming $T_e = T_i \equiv T$, one obtains for the so-called triple product¹⁷

$$nT\tau_{\text{classic}} \propto \frac{S^*}{E} L_s B T^2 \sqrt{\beta}. \quad (2)$$

Equation (2) shows that the efficiency of an ideal FRC-based fusion reactor increases with S^*/E . It is, therefore, desirable to surpass the operational boundary from Eq. (1) (namely, $S^*/E \leq 3$).

Barnes and Milroy¹⁸ showed that injection of a minority energetic ion population can provide a restoring force against tilt and, therefore, FRCs with neutral beam injection (NBI) are not limited by the tilt mode as purely thermal FRCs are. Recent TAE experimental results support this finding, where stable FRCs with S^*/E well above 3 have been recorded.¹⁹ However, as will be described below, so far simulation attempts to show stable FRC evolution with a significant fast ion population have found strong toroidal beam driven modes leading to rapid disruption. Consequently, the expected tilt stabilization through NBI has so far not been shown in 3D simulations. The reader could, therefore, ask, how is it that experimental data show stable FRC evolution (with NBI) at high S^*/E , but simulation has not been able to reproduce this?

In this paper, we resolve this long-standing discrepancy by presenting results indicating that self-consistently including magnetic evolution and NBI in simulation, results in a self-limiting mechanism that prevents the previously seen destructive toroidal beam driven modes. We hypothesize that the “micro-bursts” observed in C-2U²⁰ and C-2W is the experimental manifestation of this self-limiting cycle, where the same beam driven modes seen previously in simulation can, at low amplitude, redistribute the fast ion population in such a way to limit the mode growth while still stabilizing tilt. We then present fluid-kinetic hybrid simulation results showing stable evolution of an FRC with NBI at $S^*/E > 4$ [i.e., above the operational boundary from Eq. (1)].

The paper, therefore, has two connected foci: First, we show how stable fast ion populations can form in FRCs (Sec. IV) and second that a high S^*/E FRC with such a fast ion population can evolve stably for hundreds of Alfvén times (Sec. V). Before jumping in to new results, Sec. II gives a short review of beam driven modes in FRCs and Sec. III provides details of the models used in this work. The paper is concluded in Sec. VI with a summary and outlook for the future work.

II. BEAM DRIVEN MODES

Several authors have studied beam driven modes in FRCs, including work with analytic models, simulation, and experiment.

A. Analytic models

Finn and Sudan²¹ used an energy principle model to investigate beam driven modes. They showed beam ions can drive MHD modes if those modes are resonant with the beam (betatron²²) particles and have phase velocity in the direction of the beam current. Subsequently, this resonance condition was worked out to be^{23,24}

$$\omega^2 \approx (\omega_r - n\Omega)^2, \quad (3)$$

where ω_r and Ω are the fast ion radial bounce frequency and toroidal going-around frequency, respectively. As pointed out by Deng *et al.*,²⁰

the precession frequency, Ω_{precess} , for betatron particles with orbital character n , satisfies

$$n\Omega_{\text{precess}} = \omega_r - n\Omega. \quad (4)$$

Therefore, the resonance condition in Eq. (3) can be written as

$$\omega^2 \approx (n\Omega_{\text{precess}})^2. \quad (5)$$

For this reason, the destabilization of a rotational MHD mode due to resonance with fast ions are referred to as the “precession mode.” Note that fast ions precessing in both the ion and electron directions can resonantly excite a mode rotating in the ion direction, since in either case the particle motion is in the ion direction (either case satisfies the condition that the particle motion is in the same direction as the wave phase velocity). A sufficient condition for stability is $|n\Omega_{\text{precess}}| < |\omega|$, but precession at a significantly higher frequency than the mode will also not result in strong mode drive.

B. Experiment

Deng *et al.*²⁰ reported experimental observation of a fast ion driven instability in the C-2U experiment over a frequency range of 50–150 kHz. So-called micro-burst instabilities were seen in far-infrared laser interferometer and edge magnetic probe arrays, with clear $n = 2$ azimuthal mode number and $k_z = 0$. The mode was observed to rotate in the ion diamagnetic direction, consistent with a mode driven by the ion motion. These details, along with equilibrium field calculations and orbit tracking of fast ions (in the experimental energy range) showing beam particle orbits to primarily have $n = 2$ character, suggest the observed mode is the precession mode predicted by Finn and Sudan.²¹ Importantly, these modes were macroscopically non-destructive, causing only minor decreases in the excluded flux radius (proxy for separatrix radius) and fast ion confinement. It was found that the period between bursts was linearly proportional to the burst amplitude. This was hypothesized to be due to the higher amplitude bursts scattering more fast ions and consequently more time was needed to build back up the lost particles.

C. Simulation

Harned²³ presented some of the earliest hybrid-PIC simulations of beam driven FRCs. He expanded on the analytical work of Lovelace,²⁵ who suggested there exists a threshold in the self-magnetic field index, μ_s ,²⁶ of an energetic particle layer, above which the layer would be unstable. In particular, Harned used hybrid-PIC simulations in order to relax assumptions in Lovelace’s analysis and found that more realistic systems showed a lower threshold on μ_s to prevent instability. Above the stability threshold, growth of rotational modes in the background plasma was observed. These were explained to be due to a resonant interaction of the betatron motion (precession) with a compressional Alfvén wave (CAW) in the background plasma. The reduction in the stability threshold was attributed to the thermal spread of the energetic ions in the hybrid-PIC simulations allowing for particles with too low a precession frequency to still resonate with the background CAW. Harned observed that due to the beam driven modes in his simulations, “a large fraction of the directed beam energy is lost to thermal energy.” The precession mode prevented efficient current drive from betatron particles as the mode would scatter energy from betatron ions to thermal ions and in doing so change the orbits of the

energetic ions to Fig. 1 or drift orbits (which are not as efficient at creating azimuthal current). Similar results (fast growing modes destroying fast ion current) was seen in NIMROD,²⁷ FPIC,²⁸ and HYM²⁹ simulations as well as WarpX (see Sec. IV).

The results discussed in this section highlight a historical discrepancy between FRC simulations and experiment. Experimentally stable FRCs with a substantial populations of energetic ions are observed, but efforts to simulate such a system resulted in large amplitude azimuthal modes that drained energy from the energetic population. In this paper, we show how this discrepancy is resolved. Previously reported simulations decoupled the formation of the fast ion population from the dynamics of the FRC plasma or only accounted for axisymmetric response, i.e., a FRC equilibrium with energetic particles was obtained that did not take into account how the energetic particle distribution would self-consistently evolve with the FRC plasma as the energetic population was forming. In this paper, we show that continuous interaction between newly formed fast ions (from NBI) and the plasma results in a reorganization of the fast ion distribution from the source distribution into one that does not destabilize bulk plasma modes and thereby allow long-lived, stable, beam-driven FRCs with substantial fast ion current.

III. MODEL: KINETIC-FLUID HYBRID PIC WITH DSMC

As described in the Introduction, ion-kinetic effects play a crucial role in determining global stability in FRCs; therefore, a model with kinetic ions is needed to study FRCs. Global modes in this system occur on scales larger than the ion inertial length and frequencies lower than the ion cyclotron frequency. On these scales, electron kinetic effects can be neglected and the electrons can be treated as a continuous fluid. A fluid-kinetic hybrid model with kinetic ions and fluid electrons is, therefore, appropriate to study FRC global stability. TAE recently added such a model to the open-source particle-in-cell code, WarpX.^{30,31} A brief overview will be provided here, while a detailed description of the model and its implementation can be found in Ref. 14 as well as the WarpX documentation.³²

If one assumes quasi-neutrality (i.e., $n_e \approx n_i \equiv n$) and inertialess electrons, the electron momentum equation yields Ohm's law:³³

$$\vec{E} = \frac{1}{ne} (\vec{j}_e \times \vec{B} - \nabla \cdot \vec{P}_e) + \eta \vec{j} - \eta_H \nabla^2 \vec{j}, \quad (6)$$

where $\vec{j} = \vec{j}_e + \vec{j}_i + \vec{j}_{ext}$ is the total current density, $\vec{j}_e = -en\vec{U}_e$ is the electron current density, $\vec{j}_i$ is the ion current density, $\vec{j}_{ext}$ is the

external (non-plasma) current density,³⁴ $\vec{P}_e$ is the electron pressure tensor, n is the plasma density, η is the plasma resistivity, η_H is the plasma hyper-resistivity, and e is the electron charge. Neglecting the displacement current term in the Maxwell–Ampère equation allows one to obtain the full current density from the magnetic field

$$\mu_0 \vec{j} = \nabla \times \vec{B}, \quad (7)$$

where μ_0 is the vacuum permeability. This, in turn, allows $\vec{j}_e$ to be calculated from the knowledge of $\vec{B}$, $\vec{j}_i$, and $\vec{j}_{ext}$ via

$$\vec{j}_e = \frac{1}{\mu_0} \nabla \times \vec{B} - \vec{j}_i - \vec{j}_{ext}. \quad (8)$$

The electron pressure is taken to be a scalar and is calculated by treating the electrons as an ideal gas

$$\vec{P}_e \rightarrow P_e = n_0 T_{e0} \left(\frac{n}{n_0} \right)^\gamma, \quad (9)$$

where n_0 and T_{e0} are the reference density and temperature, respectively, and γ is the adiabatic index with value 1 for the isothermal limit and 5/3 for the adiabatic limit. The electron pressure, therefore, only depends on the local plasma density. These simplifications result in a state equation for the electric field which, along with Faraday's Law for the evolution of the magnetic field, completes the model

$$\vec{E} = \frac{1}{en} \left[\left(\frac{\nabla \times \vec{B}}{\mu_0} - \vec{j}_i - \vec{j}_{ext} \right) \times \vec{B} - \nabla P_e \right] + \eta \vec{j} - \eta_H \nabla^2 \vec{j}, \quad (10a)$$

$$\frac{\partial \vec{B}}{\partial t} = -\nabla \times \vec{E}. \quad (10b)$$

The kinetic ion particles (which determine $\vec{j}_i$ and n) are treated in the standard PIC manner and pushed using Newton's equation of acceleration and the Lorentz force (with the resistivity and hyper-resistivity terms subtracted from the E -field used in calculating the Lorentz force to conserve momentum, see Ref. 35). The standard particle advancing scheme commonly referred to as the Boris-push³⁶ is used to advance the kinetic particles. It should be noted that since the publication of Ref. 14, the B-field push has been updated to use RK4,^{37,38} which significantly improved code performance.

WarpX includes a direct-simulation Monte Carlo (DSMC) binary collision module broadly based on the kinetic–kinetic collision algorithm from Nanbu,³⁹ where particles of the colliding species are grouped into collision pairs based on their cell locations. Details of the WarpX collision module and verification tests of it can be found in Ref. 40. In particular, charge exchange collisions between a kinetic ion species and kinetic neutral species can be captured. This functionality is used in the simulations discussed here to model fast ion formation from NBI. In such cases, high energy neutral particles are injected according to beam settings (impact parameter, injection angle, etc.) and charge exchange collisions between the injected neutral particles and thermal ions in the FRC form energetic ions.

IV. STABILIZING BEAM DRIVEN MODES

In this section, we will demonstrate that the presence of an appropriate electromagnetic perturbation during the formation of the fast ion distribution in an FRC equilibrium can reorganize the fast ions in such a way that strongly reduces beam driven rotational instabilities.

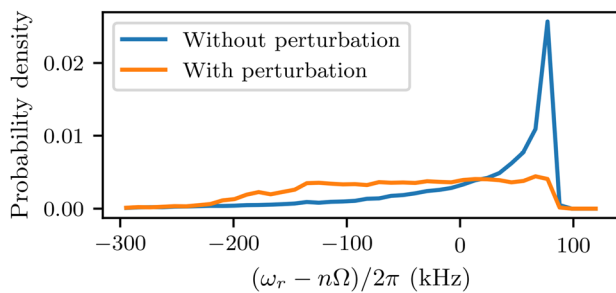


FIG. 1. Distribution of precession frequencies for $n = 2$ orbits before and after the sampled particles were pushed through an equilibrium field perturbed with Eq. (16). This case used parameters from Table I, $\zeta = 0.5$, and $A_0 = 2.5$ cm.

The general hypothesis is that the experimentally observed microbursts,²⁰ which present as low amplitude, short lived $n = 2$ excitations, perform exactly such a reorganization function in experiment, thereby allowing the build up of non-resonant fast ions while preventing the accumulation of resonant fast ions which would otherwise lead to high amplitude, destructive beam-driven modes (as seen in earlier simulation described in Sec. II). In order to clearly demonstrate the effect, we will first apply this to the same equilibrium cases considered by Harned²³ and afterwards to FRCs resembling the C-2W experiment.

A. Revisiting Harned (1982) simulations

The hybrid-PIC simulations performed by Harned²³ assumed an infinitely long FRC (2d simulations) with a cold background plasma (i.e., $T_e = T_i = 0$). An exponential rigid rotor equilibrium⁴¹ was used, for which the fast ion distribution is given by

$$f(r, v_r, v_\theta) = \frac{Mn_b}{\pi k_B T_b} \cosh^{-2} \left(\frac{r^2 - r_1^2}{r_0^2} \right) \times \exp \left(-\frac{M}{k_B T_b} \left[v_r^2 + (v_\theta - r\Omega)^2 \right] \right), \quad (11)$$

where n_b is the beam (fast ion) density, T_b is the thermal temperature of the fast ions, Ω is the angular fluid velocity of the fast ion species, r_1 is the center radius of the distribution, and r_0 is its width. The resulting magnetic field profile is

$$B_z(r) = B_{ext} - \frac{\mu_0 n_b e r_0^2 \Omega}{2} \tanh \left(\frac{r^2 - r_1^2}{r_0^2} \right), \quad (12)$$

where B_{ext} is the externally applied magnetic field strength (in the positive $\hat{z}$ -direction). The field-reversal factor, ζ , measures the strength of the B-field due to the fast ions relative to the applied field

$$\zeta \equiv \frac{|B_z^b(r=0)|}{B_{ext}} = \frac{|B_{ext} - B_z(r=0)|}{B_{ext}} = \frac{\mu_0 n_b e r_0^2 \Omega}{2 B_{ext}} \tanh \left(\frac{r_1^2}{r_0^2} \right). \quad (13)$$

Note that $\zeta > 1$ corresponds to the field-reversed condition. Harned found that the growth rate of beam-driven modes depends on the value of ζ . At low ζ values, the $n = 2$ mode is most unstable, but as ζ is increased the dominant mode number also increases. For the current purpose, we will focus on select cases from Fig. 2 of Harned's paper.²³ The parameters used to reproduce these cases are shown in Table I. Harned saw rapid growth of beam driven modes in the cases considered (his growth rates are reproduced in the simulations without a perturbation applied below).

We propose the following procedure to mimic the effect that recurring micro-bursts would have during the build up of the fast ion population: we sample fast ions according to Eq. (11) and then integrate their orbits through a static perturbed magnetic field for several cyclotron periods (40 μ s was used here, which is similar to the reported micro-burst duration in experiment²⁰). This results in a modified (perturbed) fast ion distribution, which is then evolved self-consistently (i.e., with electromagnetic field evolution) in WarpX. The perturbation we use is meant to emulate the beam driven mode that forms when evolving the unperturbed fast ion distribution coupled with the background plasma and is characterized by the displacement

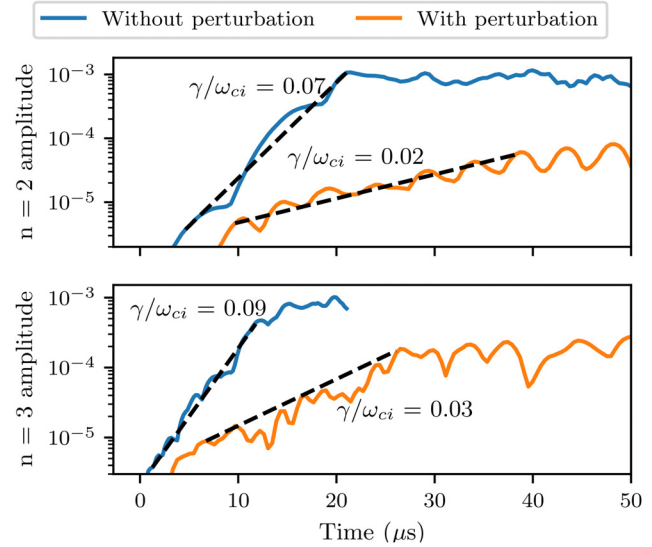


FIG. 2. Time evolution of the amplitudes of the $n = 2$ (top) and $n = 3$ (bottom) components of $\langle B_z \rangle$, for different ζ value equilibria from Harned²³ (top: $\zeta = 0.5$ and bottom: $\zeta = 1.1$), showing the evolution with fast ions directly sampled from Eq. (11) as well as when the fast ions were reorganized using the perturbation from Eq. (16). The fitted mode growth rates are also shown for each case with dashed lines indicating the linear growth period used in the fit. The cases with reorganized fast ion distributions show significantly lower mode growth rates and lower saturation amplitudes.

$$\begin{aligned} \xi_r &= A(r) \cos(n\theta) \\ \xi_\theta &= -\frac{1}{n} \left(A(r) + r \frac{\partial A(r)}{\partial r} \right) \sin(n\theta) \\ \xi_z &= 0, \end{aligned} \quad (14)$$

with

$$A(r) = A_0 \exp \left(-\frac{(r - r_1)^2}{2r_0^2} \right), \quad (15)$$

where A_0 is an input parameter that controls the perturbation amplitude. In the simulations shown here, we used $A_0 = 2.5$ cm, which results in a perturbation with 5% of the external field strength. The perturbed magnetic field is then obtained from $\delta \vec{B} = \nabla \times (\zeta \times \vec{B}_0)$, which using Eq. (12), gives

TABLE I. Equilibrium and numerical parameters used for reproduction of selected results from Harned.²³

Equilibrium parameters		Numerical parameters	
r_0	1.35 m	Number of cells	10^4
r_1	2.5 m	Fast ion	10^6
T_b	50 eV	macroparticles	
B_{ext}	0.05 T	Thermal ion	10^6
n_{plasma}	$5 \times 10^{19} \text{ m}^{-3}$	macroparticles	
ζ	0.5 or 1.1	Domain width	$5r_1$
A_0	0 cm or 2.5 cm	Timestep	$0.1/\omega_{ci}$ at B_{ext}

$$\delta B_z(r) = -\zeta_r \left[\frac{B_z(r)}{r} + \frac{2r}{r_0^2} \left(\frac{B_{ext}\zeta}{\tanh(r_1^2/r_0^2)} - (B_z(r) - B_{ext}) \right) \right]. \quad (16)$$

We note that after application of this procedure the system will no longer necessarily be in true equilibrium, since the fast ion distribution will be modified. However, this section is mainly for demonstration purpose, for which the approximate equilibrium is sufficient. In Sec. IV B, we take care to calculate a consistent equilibrium with the perturbed fast ion distribution. The effect of the field perturbation on the fast ion distribution can be seen in Fig. 1, where the probability density distribution of $n\Omega_{precess}$ is shown for a set of fast ions generated from the distribution in Eq. (11) before and after the particles interacted with the perturbed magnetic field. The perturbation smooths out the distribution of precession frequencies, reducing the accumulation of fast ions precessing at a similar frequency which would provide strong drive for a CAW in the background plasma [see Eq. (5)]. At the same time, the perturbation had very little effect on the total current produced by the fast ions (the fast ion current reduced by only 0.3%) and after the application of the perturbation all fast ions were still in betatron orbits. The impact of the change in fast ion distribution on the stability of the configuration can be seen in Fig. 2, which shows timeseries of the $n = 2$ mode amplitude for both sets of fast ions (original and perturbed), obtained from the output of evolving the initial conditions in hybrid-PIC simulations (with WarpX). The case with the smoothed out fast ions precession distribution shows substantially slower growth of the $n = 2$ mode and it saturates at a lower amplitude than the original case. Figure 2 also shows a higher ζ value case ($\zeta = 1.1$), where the $n = 3$ mode is dominant, and the fast ions were pretreated with an $n = 3$ perturbation, before performing the hybrid-PIC simulations. In that case, we again see a substantially lower mode growth rate and saturation amplitude after the perturbation was applied compared to the original equilibrium.

The results in this section show that a perturbation in the magnetic field of an FRC can cause the precession of the fast ions to change in such a way that the drive for exciting rotational MHD modes are diminished. In Sec. IV B, we apply this idea to FRC equilibria relevant to the C-2W experiment at TAE.

B. 3D FRC equilibrium

In this section, we will show that a low amplitude $n = 2$ perturbation naturally occurs as fast ions are formed from NBI and that this perturbation prevents destructive excitation of compressional Alfvén waves as has been seen in previous simulations. We show nonlinear simulations in which we evolved an FRC equilibrium that contains only thermal particles (which we refer to as a thermal FRC) while continuously injecting energetic neutral particles. The direct-simulation Monte Carlo module in WarpX was used to capture binary charge exchange collisions between neutral particles and thermal ions, capturing the source term for energetic ions. This approach goes beyond what has been reported previously since in these simulations the fast ions continuously interact with the plasma, and thereby the fast ion distribution is formed self-consistently with perturbations that arise in the electromagnetic fields.

1. Initial FRC equilibrium

An equilibrium FRC profile generated from the model described in Ref. 42 was used for the initial conditions in this study. Figure 3

shows profiles of the initial equilibrium plasma density, ion temperature, and magnetic field lines. The elliptical parameters of the separatrix in the equilibrium was $L_s = 3$ m and $R_s = 0.4$ m while the equilibrium density profile was such that $S^* = 16.1$ (giving $S^*/E = 4.3$). The external magnetic field strength was 900 Gauss close to the cylinder wall at the mid-plane, with magnetic mirrors on both ends of the central vessel with mirror ratio 5.6. The equilibrium current was fully carried by the electrons, i.e., there was no ion rotation in the equilibrium.

2. NBI parameters

Eight neutral beams were simulated, all injecting at $\pm 20^\circ$ from normal to the machine axis (z -axis in Fig. 3) with impact parameter of 20 cm; the beams were positioned such that they aimed at the machine cross section at $z = \pm 5$ cm. The beams were located symmetrically over the xy -plane (4 beams on each side), injecting toward the mid-plane with a beam located every 90° around the z -axis. The beams were set to inject at 25 keV with 100 eV thermal spread. The beam current was varied in the simulations described below.

3. Simulation parameters

A $112 \times 112 \times 288$ grid with 1.5 (3.5) cm cell resolution in x and y (z) was used for all the hybrid-PIC simulations in this section. A conducting embedded boundary (absorbing for particles) was used to mimic the vessel wall as outlined in Fig. 3. The time step was set to 0.58 ns and 20 substeps were used in updating the B-field in the RK4 scheme. The simulation was initialized with 0.4×10^9 macroparticles (~ 110 particles per cell) distributed according to the density and temperature profiles from the aforementioned equilibrium. Particle velocities were sampled from a Maxwellian distribution with the local temperature based on the equilibrium profile. NBI was simulated by injecting 250 neutral macroparticles for every neutral beam at appropriate locations at the vessel wall every 5 simulation steps with energy and velocity direction set by the aforementioned beam parameters. The weight of the macroparticles were adjusted based on the desired neutral injection rate. The Esirkepov current deposition algorithm was used with quadratic splines for the deposition and interpolation of grid quantities to reduce noise. All simulations were performed on the Perlmutter supercomputer using 2 GPU nodes, i.e., 8 A100 NVIDIA GPUs. On average, the simulations required ~ 64 h to complete the 250 μ s evolution time.

4. Hybrid-PIC evolution

Simulations were performed with different NBI current values (250 A, 500 A, and 1 kA) to investigate how the plasma response changes. The resulting build-up of azimuthal current from energetic ions for each case is shown in Fig. 4 along with a time-trace of the B_z $n = 2$ mode amplitude (volume averaged). The results show the formation of a $n = 2$ perturbation for which the growth rate increases with increased NBI current. In all three cases, the $n = 2$ mode saturates at low amplitude ($\sim 10\%$ of the vacuum field), and a clear transition in the slope of the energetic ion current build-up can be seen before and after the mode saturation. Considering that the plasma density is not substantially changed by the perturbation (i.e., no significant change to the fast ion source term), it suggests that the $n = 2$ mode increases the

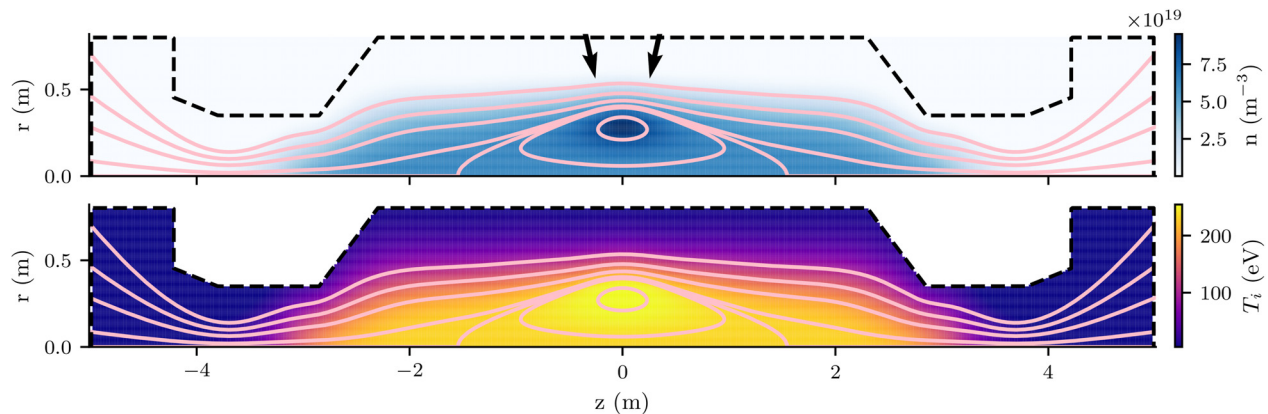


FIG. 3. Profiles of the plasma density (top) and ion temperature (bottom) of the initial equilibrium used in self-consistent simulations of fast ion formation through NBI. Traces of the magnetic field lines are also shown (solid) as well as the location of vessel walls (dashed). Arrows in the top panel indicate the location and direction of neutral beam injection (note that the injection is done with an impact parameter of 20 cm).

loss rate of energetic ion momentum in the θ direction. This matches our expectation as the mode scatters fast ions out of betatron orbits into drift orbits, which contribute less to the net energetic ion current. Nonetheless, the fast ion current continues to increase throughout the simulation, while the mode amplitude remains mostly steady at its original saturation level (there is some decay followed by regrowth cycles). The effect of the persistent low amplitude $n = 2$ perturbation on the distribution of fast ion precession frequencies is shown in Fig. 5 where a comparison is made between the fast ions formed with the self-consistent field evolution vs a reference case where the field was

not evolved during the simulation. Similar to the results discussed in Sec. IV A, the $n = 2$ perturbation smooths out the distribution of betatron precession frequencies, reducing the number of particles able to drive a CAW in the background plasma at a given frequency [see Eq. (5)]. We note that in agreement with the literature reports,¹³ in the initial phase of the simulation ion spin-up ($n = 0$ azimuthal mode) in the SOL is observed in the ion-diamagnetic direction, which is consistent with end-shortening⁴³ due to the conducting boundaries used. This resulted in the bulk SOL plasma rotating at approximately -25 kHz, and this rotation has been accounted for in Fig. 5.

The results in this section demonstrated that NBI in FRCs can induce low amplitude $n = 2$ perturbations in the magnetic field and that those low amplitude perturbations can modify the distribution of betatron precession frequencies such that the energetic ion population does not excite large amplitude CAWs in the thermal plasma as has been reported from several earlier works described in Sec. I. The low amplitude perturbations also do not prevent build-up of substantial betatron current but does reduce the rate at which such current is formed for a given NBI current. Higher NBI current was seen to

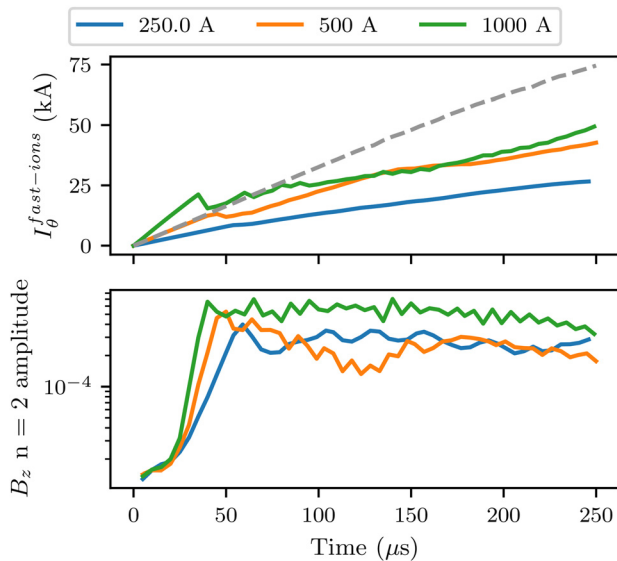


FIG. 4. Results from evolving a thermal FRC with different amounts of energetic neutral injection from NBI, showing the build-up of azimuthal energetic ion current (top) as well as the resulting growth and saturation of the $n = 2$ component of magnetic field perturbation (bottom). The dashed gray line shows the fast ion current accumulation for a comparative case with 500 A NBI but without updating the equilibrium fields during the simulation.

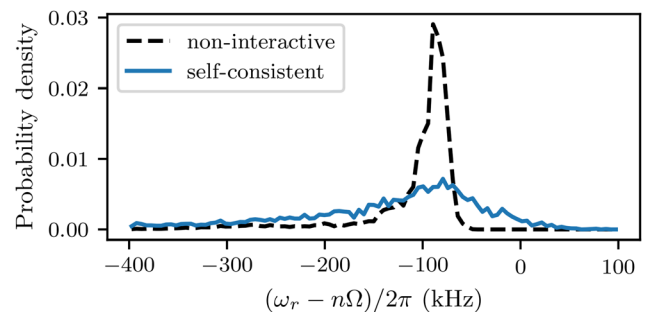


FIG. 5. Distribution of precession frequencies of betatron particles at the end of the simulation for the 500 A case shown in Fig. 4 compared to a reference case with the same beam current but where the equilibrium fields were not updated during the simulation, i.e., the betatron particles did not form with the effects of self-consistent field perturbations.

induce larger perturbations, a result that qualitatively agrees with experimental observations from C-2W.

V. ENERGETIC IONS STABILIZING TILT

Now that we have shown how self-consistently building up energetic ion current, while accounting for induced field perturbations, resolves the earlier mentioned contradiction between FRC experimental results and previously reported simulation results, we can revisit the question of whether energetic ion current can stabilize high S^*/E FRCs against tilt. In fact, the results shown in Fig. 4 already demonstrates such stabilization as the equilibrium used has $S^*/E = 4.3$ (recall the expected kinetic stability boundary is at $S^*/E \approx 3$).

The evolution of the 500 A NBI case from Fig. 4 as well as a simulation of the same equilibrium without NBI is shown in Fig. 6. Specifically, the evolution of the amplitude of the tilt mode, the separatrix radius (R_s), and the S^*/E value are shown. The amplitude of the tilt mode was calculated following the approach described in Ref. 44 (see also Refs. 45 and 14) wherein the symmetric axial ion-velocity field (over the $z = 0$ mid-plane) is calculated as follows:

$$v_z^+(\theta, z) = \int_0^{r_{\max}} (v_z(\theta, z) + v_z(\theta, -z)) dr, \quad (17)$$

and Fourier decomposed into axial and azimuthal modes. The $m = 1$ and $n = 1$ modes correspond primarily to the tilt mode, as shown in Fig. 6. The case with NBI shows saturation of the tilt mode at relatively low amplitude (compared to the saturation amplitude for the case without NBI) and the separatrix radius is seen to remain close to the equilibrium value, while the case without NBI sees the separatrix rapidly shrink once the tilt reaches high amplitude. As the case without NBI undergoes tilt and the separatrix shrinks, its value of S^*/E also

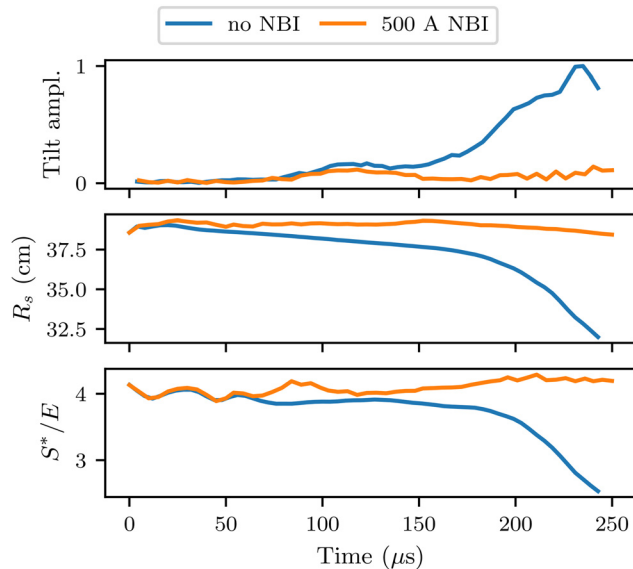


FIG. 6. Time evolution of the tilt mode amplitude (top), separatrix radius (middle), and S^*/E (bottom) for two simulations of the equilibrium from Fig. 3. In one case, NBI was added (500 A). The difference in evolution of the two cases shows tilt stabilization due to the presence of energetic ions. The tilt amplitudes were normalized such that the case without NBI has a peak tilt amplitude of unity.

decreases. In the case with NBI, S^*/E remains around 4 for the full simulation duration (in fact it increases slightly as NBI increases the total current). Figure 7 shows snapshots of the plasma density for both simulations at a late time ($t = 235\mu s$), when the no-NBI case's tilt growth saturates. The case with NBI is seen to resist the FRC tendency to tilt, even though this case has S^*/E significantly above the threshold for stabilization from thermal ion kinetic effects (as discussed in Sec. I). The large Larmor orbits of the energetic ions formed from NBI are believed to provide a restoring force against the tilt, as described in Ref. 18, which keeps the tilt amplitude low, as seen in Fig. 6.

The simulations described here did not have a source of thermal ion particles (recall that absorbing walls were used in the simulations). In all cases, the FRC density, therefore, decreases over the simulation duration. We believe the observed decrease in R_s at later times in the case with NBI from Fig. 6 is due to this slow particle loss. In future simulations, a particle source term will be added to maintain FRC density.

VI. DISCUSSION AND FUTURE OUTLOOK

In this paper, we showed how a long-standing discrepancy in beam driven FRC modeling is resolved. The discrepancy was that earlier attempts to model beam driven FRCs found destructive energetic particle induced compressional Alfvén waves (CAW), the so-called precession mode, while experimental results found only low amplitude, nondestructive modes of the same sort. The resolution to this contradiction presented here is that during the build-up of fast ion current, low amplitude perturbations lead to a redistribution of the betatron precession frequencies which avoids the excitation of the aforementioned CAW. We then showed a proof-of-principle simulation in which the presence of fast ion current stabilized a high S^*/E FRC against tilt. This result was originally predicted by Barnes and Milroy¹⁸ but has not previously been demonstrated in full 3D simulations. Future studies will explore the stability boundaries of FRCs with substantial fast ion current, aiming to understand how FRC parameters (such as S^* , excluded flux, etc.) affects the efficacy of the energetic population to stabilize the FRC against MHD modes. We will also explore how NBI settings impact the precession frequencies of the formed energetic

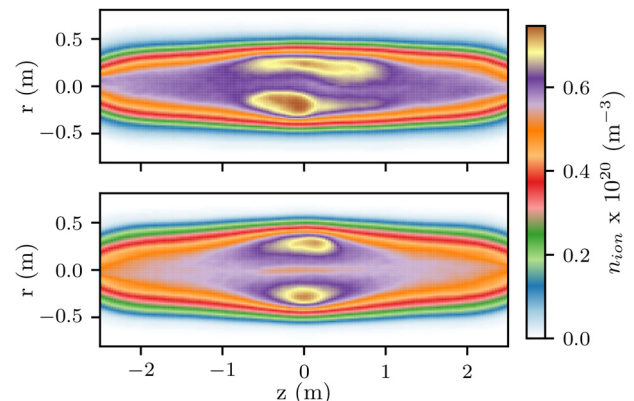


FIG. 7. Cross section of the plasma density at $t = 235\mu s$ for the two simulations shown in Fig. 6. The case without NBI (top) shows clear tilt disruption of the FRC, while the plasma structure remains intact for the case with NBI (bottom).

ions, aiming to optimize the beams for maximal current drive while avoiding the excitation of CAWs. Studying the structure of the $n=2$ perturbation that forms during the fast ion build-up, as reported here, will allow us to replicate such a perturbation and apply it as a static perturbation during fast ion formation. This would enable substantial computational savings and allow much broader sweeps of beam parameters to be studied.

A detailed comparison between results from simulation such as those described in this paper and experimental results from the C-2W experiment is under way and will be reported in a subsequent publication.

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AUTHOR DECLARATIONS

Conflict of Interest

Yes, Authors R.G., S.G., A.V., M.T., M.O., D.B., and S.D. are employees of TAE Technologies. Authors L.G. and F.C. are paid consultants for TAE Technologies. TAE Technologies is a privately funded company working toward commercialization of a nuclear fusion power plant.

Author Contributions

R. E. Groenewald: Conceptualization (equal); Data curation (lead); Formal analysis (lead); Investigation (lead); Methodology (equal); Software (lead); Validation (lead); Visualization (lead); Writing – original draft (lead). **S. Gupta:** Conceptualization (equal); Investigation (equal); Methodology (equal); Writing – review & editing (equal). **A. Veksler:** Data curation (equal); Formal analysis (supporting); Investigation (equal); Writing – review & editing (equal). **M. Tobin:** Conceptualization (supporting); Formal analysis (equal); Investigation (equal); Visualization (supporting); Writing – review & editing (equal). **L. Galeotti:** Data curation (supporting); Investigation (supporting). **M. Onofri:** Data curation (supporting); Investigation (supporting); Methodology (supporting). **F. Ceccherini:** Methodology (equal); Software (supporting); Validation (supporting). **D. C. Barnes:** Conceptualization (supporting); Methodology (supporting); Supervision (equal); Writing – review & editing (equal). **E. Belova:** Conceptualization (equal); Investigation (supporting); Supervision (supporting); Writing – review & editing (equal). **S. A. Dettrick:** Conceptualization (supporting); Funding acquisition (equal); Project

administration (equal); Supervision (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this paper are available from the corresponding author upon reasonable request and with the permission of TAE management. These data are not publicly available due to privacy restrictions.

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